

Build A Wi-Fi Cantenna

Boost your Wi-Fi network using, strangely enough, a can!

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Often believed to mean “Wireless Fidelity,” where the word “Wi-Fi” comes from remains a mystery. Still, most people agree it’s much better than using the name “IEEE 802.11b Direct Sequence.”

The “Can”tenna

A cantenna is a home-made antenna built from commonly found, empty tin cans. Why would your network need a cantenna, rather than an (inbuilt) antenna? Consider:

- ▣ They increase the range to a greater extent
- ▣ Provide high gain comparable to those of commercial antennas
- ▣ Can help spot Wi-Fi hotspots as you travel
- ▣ Commercial antennas are expensive

In other words, if you wish to have a wire-free network across distances of several metres, a cantenna is indispensable. It allows direct communication, at the same time maintaining good speed. In fact, you can enjoy a game of *Counter Strike* over the network across several buildings. Apart from these, it will aid in your being interpreted as the geek, at least in your group of friends!

What you need is what you get!

Building the cantenna is easy, provided you get the right material. Here, we will be putting together the classic Pringles cantenna. If you’ve ever enjoyed Pringles potato chips and have experienced the immense misery that accompanies seeing the empty can’s metallic bottom, don’t get upset. In fact, the chips should be consumed as

quickly as possible so that the cantenna can be built right away! Likewise, you may need to visit a radio shack for some special goods and the number of trips are proportional to how carefully you read the next few lines.

Visit your favourite supermarket and purchase a can of Pringles. The four items that follow can easily be found at a radio and electronics shop, while the next few can be obtained from a hardware store.



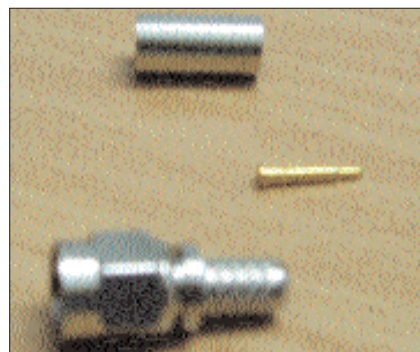
The Pringles can



The N-type female connector



The N type connector from the back



RP-SMA male connector components

You will also need

- ▣ A wire cutter
- ▣ Something to drill a hole or two
- ▣ Soldering skills

Building It

This procedure shall not take more than half an hour if you’ve already gathered the tools.



Let’s start with the can itself. The precondition is that you have consumed all the contents and properly rinsed the can. If you’re using the Pringles can, measure 3 3/8 inches from the bottom. Drill a hole such that the N type female connector can easily slide in. Now put away the can for a while.



Hole in one!

The Raw Material

Item	Quantity	Dimensions	Cost (Rs)
Pringles or equivalent can	1	Standard	80 (includes chips)
N-type female connector	1	Standard	56
RP-SMA male connector	1	Generally 8 mm dia.	76
LMR 400/RG-213 /RG-58 coaxial cable	1	1 metre	15
Copper wire piece	1	1. 1/16", Gauge 12	0
Thin conducting rod	1	6", < 1 cm dia.	12
Washers	5	Dia: inner > rod's dia; Outer = 1"	7
Small sized nuts and bolts (optional)	2	For N type connector	0